QP CODE: 25900223

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) -MG2DSCZIM102-
MOLECULES OF LIFE

Duration: 1.5 Hours **Max.Marks: 50**

Part A
Fill in the blanks
Answer **All** questions .Each question carries **1** mark

1.Glucose is a type of ……..that serve as primary source of energy.
Ans: **Monosaccharides** [K]/[CO1]

2.The two ends of a polypeptide chain are known as.....terminal and carboxy terminal. Ans: **Amino terminal** [U]/[CO3]

3.Fatty acids composed of carboxylic acid head group and long ……..chain
Ans: **Hydrocarbon** [U]/[CO4]
4.Glycolipid play a crucial role in ……..recognition and cell signalling.
Ans: **Cell-cell** [U]/[CO5]
5…… is widely used as a supporting agent to culture bacterial colonies.
Ans: **Agar** [K]/[CO1]
6…… is a digestive enzymes that plays crucial role in the digestion of maltose.
Ans: **Maltase** [U]/[CO1]
7.Globular proteins are ……..in shape.
Ans: **Spherica**l [K]/[CO2]
8.The number of grams of iodine absorbed by 100 gm of fat or oil is called…….
Ans: **Iodine number** [U]/[CO4]
9.Hemoglobin is a……protein found in red blood cells.
Ans: **Globular** [K]/[CO3]
10.The angle of rotation in peptide bond is known as……plot
Ans: **Ramachandran** [U]/[CO2]

(10 x 1 = 10 Marks)

Part B
Short answer questions
Answer any **10** questions . Each question carries **2** marks

11.Name the chemical formula of mannose and galactose and how it differ from each other. [K]/[CO1]
Ans: C6H12O6.Glucose and galactose are C 4 epimer where as glucose and mannose are C2 epimer.

12.Discuss the characteristics of globular protein. [U]/[CO2]
Ans: Globular proteins are spherical or oval in shape.,Soluble in water and other polar solvents.Most of the globular proteins have tertiary structure.

13.Give an example of a nucleotoide that serve as an energy carrier in cells. [K]/[CO5].
Ans: ATP-Adenosine Tri phosphate

14.What are acidic amino acids? [U]/[CO2]
Ans: Acidic amino acids are amino acids that have two carboxylic group in their side chains.
Eg:-Aspartic acids,Glutamic acids.

15.Outline clinical significance of PUFA. [K]/[CO5]
Ans: PolyUnsaturated Fatty acids are fatty acid with more than one double bond.They are essential for cardiovascular health,Brain and Nervous system.Eg:-Linoleicand ,Linolenic acid.

16.Explain the composition of bacterial cell wall. [U]/[CO1]
Ans: The primary component of the bacterial cell wall is peptidoglycan. Its structure is composed of N-acetyl muramic acid and N-acetyl glucosamine.

17.Discuss the role of contractile protein in human body.. [K]/[CO2]
Ans: Contractile proteins are proteins responsible for muscle contraction.
Example:- Actin,Myosin

18.Discuss the classification of fatty acids based on chain length and number of carbon atoms. [U]/[CO4]
Ans: Short chain fatty acids,Medium chain and long chain fatty acids with one example each.

19.Write short note on homopolysaccahrides. [K]/[CO1]
Ans: Homopolysaccahrides are polysaccharides that contain only one type of monosaccahride units. Example:- Starch,Glycogen

20.How did chargaff's rule contribute to the discovery of the DNA double helix. [U]/[CO5]
Ans: Fundamental principle of molecular biology.It state that the ratio of adenine to thymine is equal as the ratio of cytosine to guanine.

21.Name and explain one specific test to detect fructose. [K]/[CO1]
Ans: Seliwanoff'sTest.It is a qualitative test used to identify fructose which gives cherry red colour with this reagent.

22.Discuss the reaction and enzymes involved in transamination of amino acids. [U]/[CO2]
Ans: Transamination is a process where an amino group is transferred from an amino acids to a keto acids forming new amino acid and a new keto acids.Transaminase is the enzyme that catalyze transamination reaction.Eg;SGOT,SGPT.

(10 x 2 = 20 Marks)

. **Part C**

Short Essay Questions

Answer any **5** questions. Each question carries **4** marks

23. Explain the role of phospholipds and glycolipids in cell memberanes. [K]/[CO5]
Ans: (Any 2 importance of phospholipid and glycolipid-2 marks each)
Phospholipid-form the lipid bilayer of cell memberane,Provides memberane fluidity and permeability.Glycolipid-Contributre to cell recognition and communication,cell adhesion.

24.Differntiate between epimer and anomer. [U]/[CO1]

Ans: (Definition-1.5 marks each, Example- 1 marks)

Epimer that differ in configuration at only one specific carbon atom.Example.Glucose and mannose are C2 epimer.Glucose and galactose areC4 epimer.Anomers differ specifically at the anomeric carbon atom.Example:Alpha-D and Beta -D Glucose

25.Differntiate purely ketogenic amino acids from glucogenic amino acids. [K]/[CO2]

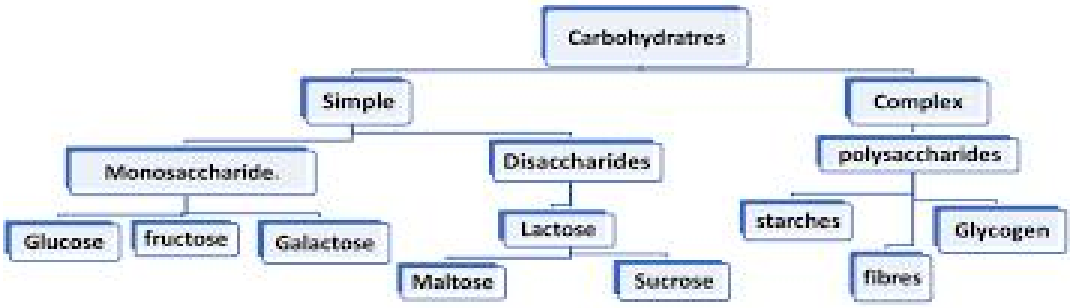
Ans: (Ketogenic amino acids with one example-2 Marks ,Glucogenic amino acids with one example-2 marks each)

Purely ketogenic amino acids are converted into ketone bodies like acetoacetate or acetyl coA and not into glucose.Eg:-Leucine,Lysine.

Purely glucogenic amino acids are converted into glucose.Eg:-Alanine.Arginine,Glycine,Histidine

26. Write an essay on different types of carbohydrates based on structure. [U]/[CO1]

Ans: (Monosaccharides -1Mark, ,Disaccharides -1 Mark,Polysaccharides-1 Mark, Structure of any one of mono or disccharides-1 Mark)



27.Explain the relationship between primary,secondary,tertiary and quaternary structure of proteins. [K]/[CO3]

Ans: (Primary , secondary,tertiary and quaternary structure-1 Mark each)

Primary structure-Linear arrangement of amino acids.

Secondary structure-alpha helix and beta pleated structure

Tertiary structure-Folded structure,Hydrogen bond,Ionic bond,hydrophobic bond-force of attraction

Quaternary structure-Subunit configuration

28.Explain the role of nucleic acid in forensic science and medical diagnostics. [U]/[CO5]

Ans: Any 4 applications - 1 mark each

29.Differentiate between starch and glycogen. [K]/[CO1]

Ans: (Mention any four differences -1Marks each)

Starch is the reserved carbohydares in plant kingdom where as glycogen isthe storage form of carbohydrates in animal kingdom.Glycogen contain a primer protein called glycogenin.Starch contain amylose and amylopectin.

(5 x 4 = 20 Marks)